Haptic Rendering Using Reality-Based Force Profiles in Surgical Simulation

Øystein Bjelland, Bismi Rasheed, Intissar Cherif, Andreas Fagerhaug Dalen, Amine Chellali, Martin Steinert, and Robin T. Bye, *Senior Member, IEEE*

***Abstract*—This paper presents a novel method for simplifying kinesthetic haptic rendering of complex contact interactions in arthroscopic surgery training simulators using reality-based force profiles. We demonstrate continuous kinesthetic feedback for applications to arthroscopic knee portal creation and diagnostic meniscus examination. This involves measuring characteristic force profiles in ex vivo experiments, simulator implementation in SOFA, and performing user validation experiments. When comparing the method with linear-elastic-based haptic feedback for meniscus stiffness discrimination, novices had difference thresholds of 1.80 MPa (linear-elastic) and 1.47 MPa (reality-based), while experts showed thresholds of 0.99 MPa and 1.39 MPa, respectively, indicating finer sensitivity among experts. Experts also used significantly less force ($p < 0.05$) and had shorter decision times ($p < 0.05$) than novices across both methods, indicating construct validity. Although kinesthetic feedback was verified with ex vivo experiments for portal creation, user validation was here inconclusive due to minor inconsistencies in the integration of visual and haptic feedback. Limitations include triggering material removal via instrument penetration instead of haptic force limits, as well as omitting contact vibrations. The method gives only a minor reduction in computation speed. Examples are available on GitHub.**

***Index Terms*—Haptic feedback, kinesthetic feedback, arthroscopic surgery, simulation.**

Received 16 September 2024; revised 4 February 2025 and 7 May 2025; accepted 14 May 2025. Date of publication 16 May 2025; date of current version 19 September 2025. This work was supported by the NTNU - Norwegian University of Science and Technology. This article was recommended for publication by Associate Editor C. Pacchierotti and Editor-in-Chief D. Prattichizzo upon evaluation of the reviewers' comments. *(Corresponding author: Øystein Bjelland.)*

This research involves the use of human cadaveric specimens, and has been considered by the Norwegian Research Ethics Committee (REC), Application 214114. The study has followed the ethical principles for medical research involving human subjects outlined in the Declaration of Helsinki.

Øystein Bjelland, Bismi Rasheed, Martin Steinert, and Robin T. Bye are with the Departments of ICT and Natural Sciences and Mechanical and Industrial Engineering, NTNU - Norwegian University of Science and Technology, 6009 Ålesund, Norway (e-mail: oystein.bjelland@ntnu.no; bismi.rasheed@ntnu.no; martin.steinert@ntnu.no; robin.t.bye@ntnu.no).

Intissar Cherif is with the Department of Computer Science, Université de Biskra, Biskra 07000, Algeria, and also with the Department of Computer Engineering, Université d'Évry, Université Paris-Saclay, 91000 Évry, France (e-mail: intissar.cherif@univ-biskra.dz).

Andreas Fagerhaug Dalen is with the Department of Research and Innovation, Ålesund General Hospital, 6017 Ålesund, Norway (e-mail: andreas.fagerhaug.dalen@helse-mr.no).

Amine Chellali is with the Department of Computer Engineering, Université d'Évry, Université Paris-Saclay, 91000 Évry, France (e-mail: amine.chellali@univ-evry.fr).

Digital Object Identifier 10.1109/TOH.2025.3570810

I. Introduction

Computer-based simulations are of increasing importance for filling the training gap needed to transition from resident doctor to specialist in surgery. Arthroscopic surgery is a field within orthopedics that is concerned with minimally invasive surgery on joints to reduce pain or restore joint function. A handful of commercial training simulators exist for arthroscopic surgery, such as Virtamed ArthroS (Switzerland), SurgicalScience ArthroMentor (Sweden) and Simendo Arthroscopy (USA). However, none of these include training in arthroscopic portal placement [1], despite being ranked as the second highest priority skill in arthroscopy [2]. The role of haptic feedback in such simulators have been described as very important. For example, in an editorial article in *Arthroscopy* from 2019, R.M. Frank concludes that *"the value of any VR simulator that does not provide tactile feedback (...) is unclear"* [3]. Moreover, in a review from 2019, Overtoom et al. assessed the value of haptic feedback in simulation training for laparoscopy [4]. They found that variable results have been found from adding haptic feedback, because existing systems cannot provide the same tactility as in fresh frozen cadavers. They also found that surgeons consider the lack of tactile feedback as an important weakness in existing simulators.

To model haptic interactions involved with arthroscopic knee surgery, probing and cutting in soft and hard tissues must be addressed. Sensory substitution through cutaneous/tactile feedback has been demonstrated for many applications within robotic surgery [5]. However, virtual reality (VR) simulations are not constrained by safety issues with stability of grounded kinesthetic feedback, as no patient harm is risked. Rather, we argue that functional task alignment, which refers to whether training is transferable to the operating room, is best preserved through kinesthetic haptic feedback. This is because cutaneous/tactile stimuli are also naturally conveyed through kinesthetic feedback, and are representative for arthroscopy, as the surgeon mostly uses an instrument to manipulate tissue. Such functional tasks can include triangulation, which involves proprioception, handling specialized surgical instruments, which require the development of dexterous motor skills, and learning the tactile sensation of tissues through a surgical instrument. For suspected meniscus injury, the surgeon creates lateral and medial incisions and then introduces a camera and a probe into the knee compartment for examination. Two tasks relying on dexterous motor skills and tactility are (i) creating incisions using a scalpel blade, where an overshoot could injure the ligaments, menisci and cartilage tissues, all of which have little healing potential, and (ii) assessing the stiffness of the meniscus tissue by probing, as a decrease in stiffness has been found to indicate meniscus degeneration [6]. A degenerative meniscus could cause arthrosis, and therefore the stiffness of the tissue could affect the decision of whether to resect or suture a meniscus tear. Convincing haptic display is, therefore, essential for safely practicing these skills. Following general triangulation, portal creation, and examination have been identified as the most important skills for simulation training in arthroscopy [2].

Our work focuses on haptic modelling of complex contact interactions experienced in surgery. We propose a reality-based method for enhancing kinesthetic feedback in a three-dimensional (3D) finite element (FE) simulation environment by enabling the use of empirically measured characteristic force profiles. The method is demonstrated for applications in the human knee, with arthroscopic portal creation and meniscus examination as examples. However, the method is applicable to any type of surgery.

II. Related Work and Contributions

A. Related Work

Computing mechanical forces from interactions between a virtual surgical instrument and a biomechanical tissue model lies at the core of this work. The computations are on the biomechanical physics side constrained by the need to satisfy visual refresh rate requirements and perform collision detection, in the range of 25–30 Hz. The mass-spring method (MSM) was previously popular due to its low computational load. However, it is difficult to achieve a realistic force response from tuning individual springs. Position-based dynamics (PBD) works directly on positions to solve geometric constraints, which makes it fast and free of instabilities. It has recently been applied for surgical simulations in laparoscopy and endoscopy [7]. However, since stiffness depends on the time step, it is challenging to achieve a realistic force response [8]. The finite element method (FEM) is considered the gold standard in soft tissue biomechanics because of its high accuracy force response, even if it has a higher computational load than PBD and MSM [9].

On the haptic side, computations need to meet refresh rates of about 500–1000 Hz to satisfy stability of the haptic device [10], as well as the sensitive tactile receptors of individuals [11]. To meet stability requirements in kinesthetic display of rigid bodies, virtual coupling, and the God-object method have been introduced [12]. These ensure visual non-penetration as well as stiffness and damping parameters that can be tuned to ensure stability and model rigid body haptic interactions. In knee arthroscopy, this is relevant for bony structures where geometry, friction and texture dominate the haptic sensation. Constraint-based methods have later been developed to handle haptic rendering of complex contact interactions in the context of FE-based soft-tissue simulations. Relevant tissues in the knee include the skin, ligaments, and menisci. Here, the contact response is formulated as the linear complementary problem (LCP) for unilateral contacts [13] and bilateral contacts [14]. This method is referred to as constraint-based multi-rate compliant mechanisms, and contact forces are computed using Lagrange multipliers. Because constraints are formulated in a physics loop and resolved in a separate haptic loop, haptic refresh rates are ensured. Such implementation exists in the Simulation Open Framework Architecture (SOFA) [15].

Multi-rate compliant mechanisms provide an elegant way of computing haptic forces from FE-based soft-tissue deformation. However, the method is limited by constitutive biomechanics inherent in FEM [16]. For example, the co-rotational FEM (CFEM) is a neat way of modelling geometric nonlinearities while using linear-elastic material parameters, thereby not having to recompute the stiffness matrix at each time step which reduces computational loads. However, the contact force response from material deformation is limited to be linear, while many interactions are nonlinear. For example, have both skin and meniscus tissue been shown to exhibit nonlinear and anisotropic behavior [17], [18], which could affect both arthroscopic portal creation and meniscus probing exercises. Although FEM with hyperelastic constitutive models can describe material nonlinearities, stability can be compromised under the constraint of computational speed which makes it risky for interactive real-time simulation [19]. Also, model fitting can be complex. Furthermore, it has been shown that coarse FE-meshes used to satisfy real-time requirements affect the accuracy, and therefore, the same mechanical contact force response as used for calibration is not guaranteed [20].

Moving into soft tissue topology change, such as cutting or material removal, kinesthetic force feedback becomes even more challenging. Some work has addressed bone removal scenarios but is limited to rigid objects [21]. With FEM, topology change is modeled with element removal and subsequent update of the global stiffness matrix, of which the updated compliance of the soft tissue can be rendered [22], [23]. This is useful for example in displaying the compliance of a cut skin. The cutFEM [24] and co-rotational cutFEM have been proposed for modelling soft tissue cutting [25]. The idea is to use a non-conforming background mesh to reduce the computational burden of mesh generation. However, a fundamental challenge in haptic rendering during tissue removal is that when an element is removed, in what we must assume is a somewhat coarse computational mesh, impedance is temporarily removed and dynamic amplifications are introduced once the instrument collides with the next element, which could compromise stability. Also, the normal directions of the cutting surface can vary quite dramatically for coarse tetrahedral meshes, making constraint-based haptic feedback unsuitable.

Pioneering work modelled haptic interactions directly from tool-tissue interaction experiments in *reality-based modelling* [26], [27]. Today, many researchers still rely on reality-based methods to develop haptic feedback for surgical procedures. However, studies often involve specialized devices that are mostly not integrated with established simulation frameworks that model 3D soft tissue physics due to cumbersome implementation [1], [28], [29], [30]. Furthermore, reality-based modelling has renewed relevance due to recent technological developments. In surgery, such as computer-assisted orthopedic surgery (CAOS), the use of intraoperative navigation and force registration techniques are making their way into the operating room, making interaction data more available than ever [31]. Data-driven methods, for example, within machine learning, have also shown good performance in modelling haptic interactions [32]. As such, there is a need for structured methods to leverage haptic rendering from FE-based simulations using reality-based methods.

B. Contributions

Motivated by the efforts of Okamura et al. [27] and Peterlik et al. [14], the primary objective of this work is to propose, demonstrate, and validate a method for reality-based enrichment of haptic interactions in the context of constraint-based kinesthetic feedback. It addresses the need for displaying complex nonlinear contact forces experienced during surgery, as well as the fundamental challenge of proving a continuous cutting force during soft tissue removal in coarse FE-meshes. We argue that the main benefit of this method is that it simplifies the focus on critical haptic interactions in the development of interactive surgical simulators, enabling safe training of dexterous motor skills. Instead of comprehensive material parameter tuning, characteristic contact forces can be used directly whether it has a linear, nonlinear, or discontinuous shape.

Our method is based on mapping 3D contact forces, measured experimentally in an arbitrary reference frame, into a one-dimensional (1D) characteristic force profile. We refer to this as a *reference force*, even though it consists of a force- and displacement signal. Although we have applied a similar principle in [33] to estimate the elastic moduli of the meniscus, our novel contribution is to use this 1D reference force directly in a 3D reality-based haptic rendering scenario. Specifically, the experimental reference force (e.g. measured by a specialized device [34]) can be rendered by extracting constraint directions from the simulation contact forces (see Sections III-A and III-B), and then combine the reference force and simulation contact force into a new haptic force signal. A method for performing this fusion using a Kalman filter has been introduced in [30], but has not been demonstrated in a full 3D simulation environment. We also present a practical contribution to transmit these forces to a kinesthetic device with three degrees of freedom (DOF) force feedback (see Section III-E).

We demonstrate our method in the case of arthroscopic portal creation in the knee joint, in which a reality-based discontinuous haptic force is displayed during cutting, as well as for arthroscopic meniscus examination, where an empirical nonlinear haptic force is enabled without the need for hyperelastic constitutive models. The method is implemented using the SOFA framework [15] (see Section V). Reference force signals are obtained from ex vivo arthroscopic knee surgery experiments (see Section IV), and a secondary objective is to describe characteristic force profiles from arthroscopic portal creation and meniscus examination.

Finally, a user validation study (see Section VI) is presented as another contribution, where we compare the performance of experts and novices using our method with a conventional linear-elastic haptic force. In addition to validating our method, we investigate whether the shape of the haptic force signal affects user performance in meniscus stiffness discrimination. We show that experts have a finer sensitivity than novices in discriminating meniscus stiffness for both methods, indicating construct validity and validating our method. We also show that there is no statistical difference between a linear and nonlinear force curve in meniscus stiffness discrimination. For arthroscopic portal creation, we find no statistical difference in scalpel insertion length between experts and novices. This is discussed in Section VII.

III. Method: Reality-Based Kinesthetic Rendering

A. Extracting Simulation Contact Forces

Our method begins by extracting contact forces from the interaction between an instrument and an FE-based deformable or rigid object, as described in [14]. A short summary is provided for reference. CFEM is used in our method demonstration because it can handle geometric nonlinearities of tissue deformations. This is needed to produce accurate contact directions as well as a convincing visual representation.

The two types of contacts concerned are *holonomic constraints*, $\mathbf{\Phi}$, meaning bilateral interaction laws such as attachment and sliding, and *non-holonomic constraints*, $\mathbf{\Psi}$, meaning unilateral interaction laws such as contact and friction. These can be denoted

$$\mathbf{\Phi}(x_1, x_2, \ldots) = 0, \tag{1}$$

$$\mathbf{\Psi}(x_1, x_2 \ldots) \geq 0, \tag{2}$$

where $x_1, x_2, \ldots$ are the objects involved with the constraint. In our method, we primarily address non-holonomic constraints, but holonomic constraints are included in the description for completeness. Assuming the deformation of soft tissue is modelled using CFEM with implicit time integration, we get the following general description:

$$\underbrace{\left(\mathbf{M} - dt\mathbf{B} - dt^2\mathbf{K}\right)}_{\mathbf{A}} \mathbf{\Delta v} = \underbrace{dt \cdot \mathbf{f}(\mathbf{x}(t), \mathbf{v}(t)) + dt^2\mathbf{K}\mathbf{v}(t)}_{\mathbf{b}} + dt\mathbf{H}^T\lambda. \tag{3}$$

where $\mathbf{M}$ is the mass matrix, $\mathbf{K}$ is the stiffness matrix, $\mathbf{B}$ is the damping matrix, $\mathbf{f}(\mathbf{x}(t), \mathbf{v}(t))$ is the sum of internal and external forces not associated with mechanical interactions, dt is the time step and $\mathbf{v}$ is the velocity. The term, $\mathbf{H}^T\lambda$, gives the nodal forces to be applied to compensate for collision. For two colliding objects, we have

$$\mathbf{A_1 \Delta v_1} = \mathbf{b_1} + dt\mathbf{H_1}^T\lambda \tag{4}$$

$$\mathbf{A_2 \Delta v_2} = \mathbf{b_2} + dt\mathbf{H_2}^T\lambda. \tag{5}$$

This is usually solved indirectly, by first computing the free motion (no constraints), and then minimizing interpenetration, as described in [14].

From the interaction laws, the Jacobian, $\mathbf{H}$, of the constraints can be computed:

$$\mathbf{H_1}(x) = \left[\frac{\partial \mathbf{\Phi}}{\partial x_1}; \frac{\partial \mathbf{\Psi}}{\partial x_1}\right]; \mathbf{H_2}(x) = \left[\frac{\partial \mathbf{\Phi}}{\partial x_2}; \frac{\partial \mathbf{\Psi}}{\partial x_2}\right] \tag{6}$$

This Jacobian represents the direction of the constraints, while the forces are computed using the Lagrange multipliers, λ, in constraint space. From here, the contact forces in cartesian space, $\lambda(x, y, z)$, can be extracted from the contact forces in constraint space, $\lambda(n, t_1, t_2)$:

$$\lambda(x, y, z) = \mathbf{H}^T \lambda(n, t_1, t_2) \tag{7}$$

where n is the force intensity in the normal direction and t_1, t_2 are the force intensities in the two tangential directions.

B. Decomposition of Contact Force and Projection of Displacement

The contact force signal from the simulation in cartesian space, as well as the experimentally measured *reference force*, is next decomposed, and mapped from three to one dimension. This is done to (i) extract the contact force direction from the simulation and (ii) prepare the simulation- and reference forces (and displacements) for reference-frame-free signal processing. The experimental reference force follows the same procedure for force decomposition and displacement projection as the simulation contact force, but we limit the explanation to the latter here.

Consider that for non-holonomic constraints, the force vector will consist of a dominant direction. For example, this would be nearly perpendicular to the surface for frictionless contact. The contact force, $\lambda(x, y, z)$, can then be decomposed into magnitude, $|\lambda|$, and direction $\mathbf{u}_\lambda$, so that:

$$|\lambda| = \sqrt{\lambda_x^2 + \lambda_y^2 + \lambda_z^2} \tag{8}$$

$$\mathbf{u}_\lambda = \frac{\boldsymbol{\lambda}}{|\lambda|}. \tag{9}$$

If needed, any of the normal or tangential directions could be selected instead ($\lambda(n, t_1, t_2)$).

Characteristic haptic interactions are most generally described by ordinary differential equations, of which reaction force is computed from a linear- or nonlinear position-, velocity- or acceleration dependent relationship. Therefore, the relevant states of a simulation target object, $\boldsymbol{\delta}$, need to be extracted. The most relevant target object will in many cases be the haptic interaction point (HIP), of which could be attached to a virtual surgical instrument model. We use this for our explanation. However, we emphasize that states of the nodes in the deformable object involved with the contact also could serve as target objects.

Because the length of the force direction unit vector is one by definition, $|\mathbf{u}_\lambda| = 1$, the projected states, $\delta(\mathbf{x}, \mathbf{y}, \mathbf{z})$ along the direction of force, $\mathbf{u}_\lambda$, can be calculated using the vector dot product:

$$|\delta_\lambda| = |\delta(\mathbf{x}, \mathbf{y}, \mathbf{z})||\mathbf{u}_\lambda| cos(\theta) = \delta(\mathbf{x}, \mathbf{y}, \mathbf{z}) \cdot \mathbf{u}_\lambda. \tag{10}$$

We are interested in the projected states because these will have the same direction as the contact force and the force-state relationship can be described. For example, a unidirectional force-displacement relationship for meniscus probing can be modelled in 3D-space, as shown in Fig. 1. This produces a meaningful 1D representation of the force-state relationship that can be used for haptic modelling.

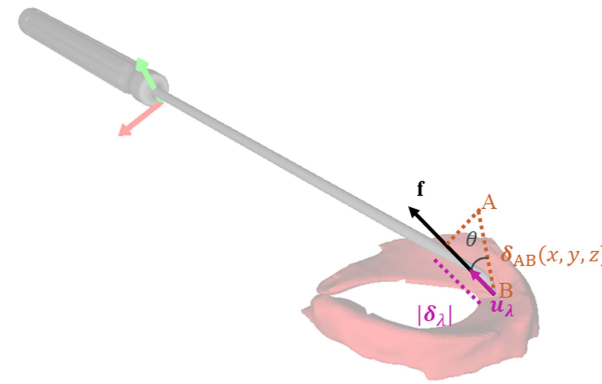

Fig. 1. As the instrument tip moves from point A to point B, the direction of the resulting force vector for non-holonomic constraints will consist of a dominant direction which is determined from the topology. Because the length of the force direction unit vector is one, the projected states $\delta(\mathbf{x}, \mathbf{y}, \mathbf{z})$ can be calculated using the vector dot product, as shown in (10).

C. Generating a Reality-Based Characteristic Force Signal

The purpose of mapping the 3D representation of the contact force to 1D is to prepare it for intuitive signal fusion or replacement by an experimental characteristic reference signal. In the reality-based haptic modelling framework, Okamura et al. describe three main categories. These are (i) physics-based models, such as ordinary differential equations, (ii) empirical models, which could be a useful mathematical representation without physical meaning, and (iii) a database of recorded interactions, which are normally rendered by means of interpolation methods [26]. Machine-learning based methods would often fall into the second category. The goal of our method is to accept characteristic force signals, $\mathrm{f}_{\mathrm{ref}}$, generated from any of these methods. In this study, we consider an empirical reference force measured by a force-torque sensor (FS), f_{FS}, and position sensor (PS), δ_{PS}, located in the surgical instrument. This reference force is first mapped to 1D using equations (8)–(10), or by selecting the dominant direction. A simulation reference force, $\mathrm{f}_{\mathrm{ref}}$, can then be generated e.g. using linear interpolation, so that:

$$\mathrm{f}_{\mathrm{ref},k} = \mathrm{f}_{\mathrm{FS},k} + (\mathrm{f}_{\mathrm{FS},k+1} - \mathrm{f}_{\mathrm{FS},k}) \frac{\delta_k - \delta_{\mathrm{PS},k}}{\delta_{\mathrm{PS},k+1} - \delta_{\mathrm{PS},k}}, \tag{11}$$

or through an empirical model, such as shown in (23).

D. Signal Fusion

To provide a smooth fusion of the experimental reference- and contact force signals, a Kalman filter observer is used. Initial efforts on this method have been described in our previous work [30], but have not been demonstrated in a 3D simulation scenario. We emphasize that this step is optional, and it is possible to simply replace the simulation force signal by the reference force signal. However, the advantage of using a Kalman filter for signal fusion is that a somewhat "rough" characteristic reference force signal can be used to generate a smooth haptic force signal.

Algorithm 1: Kalman filter signal fusion.

1: Take measurement of CFEM forces, $|\lambda|$.
2: Assign uncertainty to CFEM force measurement, R_λ.
3: Perform prediction and correction based on λ.
4: Take measurement of reference signal, $|\mathbf{f_{ref}}|$.
5: Assign uncertainty to reference signal measurement, $R_{\mathbf{ref}}$.
6: Perform new prediction and correction based on $|\mathbf{f_{ref}}|$.

A mathematical state-space representation of the haptic force signal is formulated, and the haptic force, f, and impulse, j, at time step k are selected as states. The state space representation leads to the system matrix shown in (12).

$$\mathbf{A} = \begin{bmatrix} 1 & dt \\ 0 & 1 \end{bmatrix} \tag{12}$$

As such, the a priori state estimate at time step k is

$$\hat{\mathbf{x}}_k^- = \begin{bmatrix} \hat{\mathrm{f}}_k^- \\ \hat{\mathrm{j}}_k^- \end{bmatrix} = \begin{bmatrix} 1 & dt \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\mathrm{f}}_{k-1} \\ \hat{\mathrm{j}}_{k-1} \end{bmatrix}. \tag{13}$$

As the force signal is sampled by alternating between setting the measurement at time step k, $\mathbf{z}_k$, equal to $|\lambda|$ or $|\mathbf{f_{ref}}|$, the measurement matrix is taken as shown in (14).

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{14}$$

The discrete Kalman filter, as presented in [35], is then implemented using the prediction equations shown in (15) and (16).

$$\hat{\mathbf{x}}_k^- = \mathbf{A}\hat{\mathbf{x}}_{k-1} \tag{15}$$

$$\mathbf{P}_k^- = \mathbf{A}\mathbf{P}_{k-1}\mathbf{A}^T + \mathbf{Q} \tag{16}$$

Here, $\hat{\mathbf{x}}_{k-1}$ is the a posteriori estimate at time step $k-1$, $\mathbf{P}_k^-$ is the a priori estimate error covariance at time step k, $\mathbf{P}_{k-1}$ is the a posteriori estimate error covariance at time step $k-1$, and $\mathbf{Q}$ is the system covariance matrix.

The measurement update equations are shown in (17), (18) and (19), where $\mathbf{K}_k$ is the Kalman gain, $\hat{\mathbf{x}}_k$ is the a posteriori state estimate, $\mathbf{P}_k$ is the a posteriori estimate error covariance at time step k, $\mathbf{I}$ is the identity matrix, and the rest as previously defined.

$$\mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}^T(\mathbf{H}\mathbf{P}_k^-\mathbf{H}^T + \mathbf{R})^{-1} \tag{17}$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_k^-) \tag{18}$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \tag{19}$$

The process noise covariance matrix, $\mathbf{Q}$, and the measurement noise variances associated with the simulation, R_λ and reality-based signal $R_{\mathbf{ref}}$, must be set in advance. They are here assumed to be constant. The tuning of these parameters will determine how much the reference force signal is trusted, and thereby how much of it should be weighted in the haptic force signal formation. They will also influence how much noise is removed. The signal fusion is performed as shown in Algorithm 1.

E. Transmitting Force to Haptic Device

After the new haptic force signal magnitude, f, has been generated either from (i) substitution of the magnitude of reality-based reference force signal, $|\mathbf{f_{ref}}|$, or (ii) by signal fusion (see Section III-D), a new haptic force vector is generated from f and the simulation contact force direction vector, $\mathbf{u}_\lambda$. We here present a practical contribution for transmitting these forces to a kinesthetic haptic device.

For contacts not involving topology change, the haptic force vector is generated by multiplying the haptic force signal magnitude with the contact force direction vector given at the current time step, so that

$$\mathbf{f}_k = \mathbf{u}_{\boldsymbol{\lambda},k}\mathrm{f}_{\mathrm{ref},k}. \tag{20}$$

For contacts involving topology change, a problem arises that the contact force direction can potentially vary quite chaotically from one time step to the next. In the case of tetrahedral elements, this is because collision detection algorithms base the contact direction on the new normal directions of the element surface. For coarse tetrahedral meshes, the direction changes can be large. To overcome this, the initial contact force direction vector before elements are removed, $\mathbf{u}_{\boldsymbol{\lambda}}^0$, is used so that:

$$\mathbf{f}_k = \mathbf{u}_{\boldsymbol{\lambda}}^0\mathrm{f}_{\mathrm{ref},k}. \tag{21}$$

This provides a justified representation for straight cuts. For more intricate material removal operations, such as arthroscopic meniscus shaving, the opposing direction of motion of the haptic interaction point should be used instead, so that:

$$\mathbf{f}_k = \frac{\boldsymbol{\delta}_k}{|\boldsymbol{\delta}_k|}\mathrm{f}_{\mathrm{ref},k}. \tag{22}$$

The force vector is finally sent to the haptic device driver for haptic display.

IV. Data Collection Experiments: Ex Vivo Knee Arthroscopy

A. Overview

This section describes experiments measuring characteristic tool-tissue interactions for portal creation and meniscus examination in arthroscopic knee surgery. Ex vivo human cadaveric knee joints were used. The research was approved by the Regional Ethics Committee under reference 214114. The use of cadaveric specimens followed the principles underlined in the Declaration of Helsinki. While we have previously obtained elastic moduli of the menisci from the same knee specimens in [33], this section presents novel contribution that describe force characteristics from arthroscopic portal creation and meniscus probing, measured by a force- and position sensor located in the surgical instrument.

B. Specimen Sourcing and Preparation

Five cadaveric knee joints were obtained (Science Care, USA) with a mean age of 71.8 years (68–78). The specimens, including anatomical mid-femur to mid-tibia and fibula, were sourced from two right knees (RK1, RK2) and three left knees (LK1,

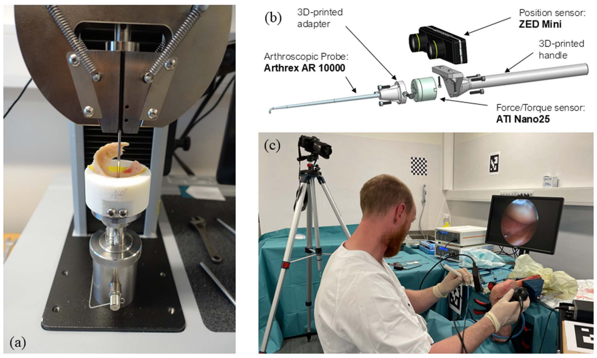

Fig. 2. Apparatus overview for ex vivo experiments. (a) Biomechanical indentation testing of meniscus specimens using the Instron 34SC-5 test system. (b) CAD-model of arthroscopic tracker instrument, including ATI Nano25 F/T sensor, and ZED Mini stereo camera. The picture shows the setup for the arthroscopic probe. For portal creation, a scalpel holder and Swan-Morton Scalpel blade no.11 were used. (c) The arthroscopic data collection experiment setup. Cadaveric knee specimens were attached to a bench using a vice. The arthroscopic tracker instrument was used to measure surgeon interaction forces and instrument position.

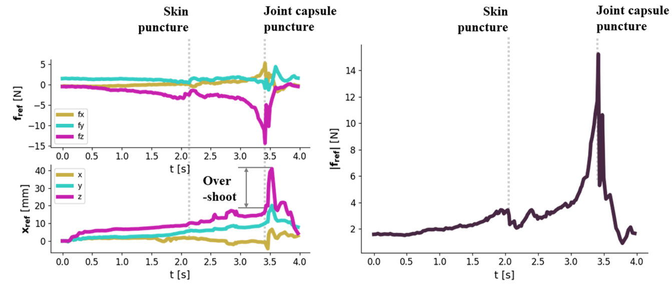

Fig. 3. Force signal components (x, y, z) for arthroscopic portal creation in LK1, and the corresponding force signal magnitude (right). The locations of skin puncture and joint capsule puncture are marked.

LK2, LK3). The donor cohort consists of three males and two females, with a mean body mass index of 26.6 (17–33). Prior surgeries on knee joints were considered as an exclusion criteria during procurement. The knee joints were stored in a freezer at -28^{o}C and were thawed at room temperature for 24 hours before arthroscopy.

C. Test Setup and Procedure

An arthroscopic tracker instrument was used to measure contact forces and instrument pose [34]. The reported accuracy was ± 1 mm for position, and 0.5° for orientation. The sampling rate was about 38 Hz. The force readings were bias-compensated and then transformed from the sensor coordinate system to the global coordinate system using the pose data. For meniscus probing experiments, position readings were transformed from the camera coordinate system to the instrument tool tip coordinate system.

The arthroscopic surgery was performed by an experienced orthopedic surgeon. For each knee, a lateral and medial portal was established using a scalpel, and the force and pose data was saved to a CSV file. Next, a trocar and arthroscope were inserted. The menisci (lateral and medial) were then examined by two probings on the superior surface in the posterior, mid, and anterior regions. This force and pose data was saved to a separate CSV file. No water was injected into the knee. The procedure, including the arthroscopic camera view, was video recorded. The different phases of the procedures could be verified by cross-checking the time stamp in the video with the measured force-pose data.

After arthroscopy, the menisci were extracted and biomechanical indentation experiments were performed using an Instron electromechanical testing system (model 34SC-5) to identify the elastic modulus for each meniscus specimen, as described in [33]. An apparatus overview is shown in Fig. 2.

TABLE I
The Lateral Portal Skin Puncture Force (LSPF), Lateral Portal Joint Capsule Puncture Force (LJCPF), Medial Portal Skin Puncture Force (MSPF) and Medial Portal Joint Capsule Puncture Force (MJCPF)

	RK1	LK2	LK3	RK2	LK1	Mean
LSPF [N]	1.50	2.26	1.51	3.13	3.22	2.32
LJCPF [N]	5.85	11.48	6.67	10.50	-	8.63
MSPF [N]	1.33	2.85	1.51	2.80	3.42	2.38
MJCPF [N]	4.52	4.11	8.29	10.62	15.24	8.56

TABLE II
The Mean Peak Force, $\overline{|\mathbf{f_{peak}}|}$, the Mean Elastic Modulus, $\overline{E}$, and Mean Poisson's Ratios, $\overline{\nu}$, for the Anterior, Mid and Posterior Regions of the Lateral and Medial Menisci

Meniscus Region	$\overline{\lvert\mathbf{f_{peak}}\rvert}$ [N]	$\overline{E}$ [MPa]	$\overline{\nu}$
Lateral anterior	8.47 ± 2.25	3.58 ± 2.76	0.30
Lateral mid	10.69 ± 2.55	2.68 ± 1.36	0.28
Lateral posterior	10.23 ± 3.47	4.24 ± 1.89	0.29
Medial anterior	11.38 ± 4.05	5.05 ± 1.97	0.28
Medial mid	11.26 ± 3.42	2.42 ± 1.68	0.26
Medial posterior	12.60 ± 3.80	4.06 ± 1.80	0.26
Average	10.77	3.67	0.27

The mean peak force was measured with the arthroscopic probe, and the elastic modulus and poisson's ratios were measured during the biomechanical indentation tests.

D. Results

1) Portal Creation: A characteristic force curve of arthroscopic portal creation from the medial side of left knee 1 (LK1) is shown in Fig. 3. The portal creation consisted of (i) skin tissue deformation, (ii) skin puncture, (iii) joint capsule deformation, (iv) joint capsule puncture. For the lateral portals, the mean skin puncture force (LSPF) was 2.32 N, and the mean joint capsule puncture force (LJCPF) was 8.63 N. For the medial portals, the mean skin puncture force (MLSPF) was 2.38 N, and the mean joint capsule puncture force (MJCPF) was 8.56 N (see Table I). The length of the the arthroscopic scalpel blade was 40 mm.

2) Meniscus Probing: Fig. 4 shows the resulting force signal from arthroscopic probing of the medial meniscus in left knee 1 (LK1). The peak forces measured in each of the respective regions (anterior, mid, posterior) of the menisci (lateral, medial) of the five knees are shown in Table II. In addition, the corresponding elastic moduli ($\overline{E}$ and Poisson's ratios, $\overline{\nu}$), from [33] are included for comparison. Fig. 5 shows the region-wise differences of the characteristic nonlinear force-displacement curve measured by biomechanical indentation.

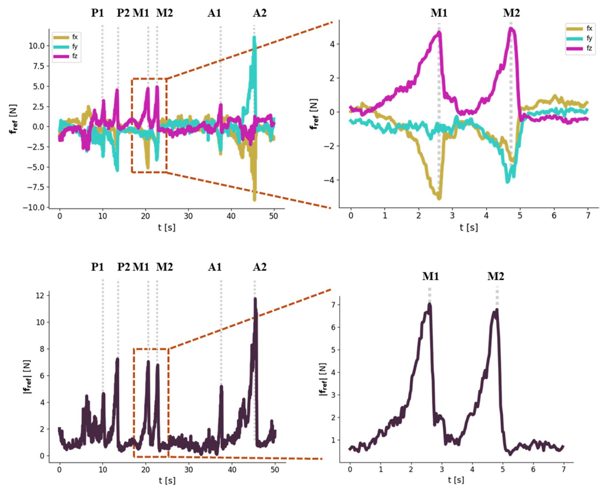

Fig. 4. Force signal components (x,y,z) for arthroscopic meniscus probing of the medial meniscus in left knee 1 (LK1) (top), and the corresponding force signal magnitude (bottom). The protocol was two probings in the posterior region (P1, P2), two in the mid-region (M1, M2), and two in the anterior region (A1, A2). The characteristic force peaks are indicated. A close-up of M1 and M2 is shown to the right.

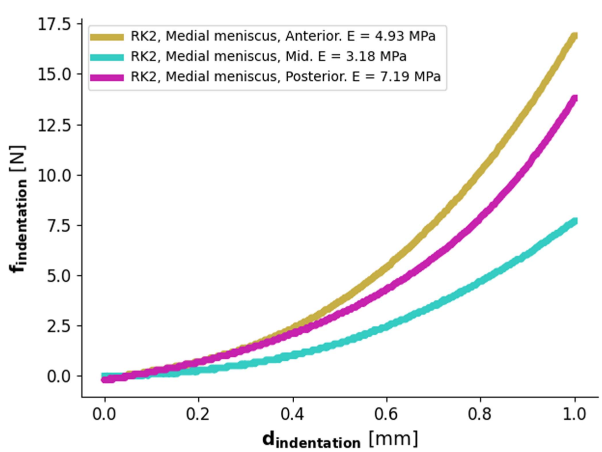

Fig. 5. The characteristic force-displacement curve for 1 mm indentation of meniscus tissue. Curves for the anterior, mid, and posterior regions are shown, as well as the resulting identified elastic moduli.

V. Surgical Simulation Implementation: Knee Arthroscopy in SOFA

A. Biomechanical Simulation Model and Haptic Setup

A biomechanical simulation model of the knee joint was developed in the Simulation Open Framework Architecture (SOFA) [15]. Anatomical models were based on magnetic resonance imaging models that were automatically segmented [36]. Computation models were modeled in CAD (Siemens NX, Germany) using MRI models as a template, and subsequently meshed using linear tetrahedral elements in GMSH [37]. The mesh density was tuned to meet real-time requirements of 30 Hz for the physics thread on a laptop (11th Gen Intel Core i5-11300H @ 3.10 GHz CPU, 8 GB memory). Surface meshes extracted from the computation models were used as collision models. For the lateral meniscus, the computation model consisted of 1263 tetrahedral elements, the collision model 512 vertices, and the visual model 11722 vertices. For portal creation, the skin model consisted of 342 vertices for the visual and collision models, and 946 tetrahedral elements for the computation model. Other visual models present in the portal creation scene made up a total of about 1.7 million vertices.

Surgical scalpel and arthroscopic probe instrument models were developed in CAD, and modeled as rigid bodies. The haptic interaction point and virtual instrument model were connected with springs, but the spring stiffness (translational and rotational) was set very high to avoid disturbance when studying contact forces. Because the tool-tissue interactions were generally for soft tissues, this was unproblematic for stability. A Geomagic Touch haptic device (3D Systems, USA) was used for haptic display. The Geomagic plugin was used in SOFA for handling device kinematics and actuation. SofaPython3 was used for accessing simulation variables and computing haptic forces. The SofaPython3 plugin exposes C++ components available in SOFA to Python. The haptic interaction point was selected as the target object, δ, and displacement was selected as the state.

B. Application to Arthroscopic Portal Creation

To model arthroscopic portal creation, tissue removal and haptic force modelling are needed. The skin was modeled as a deformable object using CFEM, with $E = 15\,\text{MPa}$. Other anatomical models were included as visual representations only. Topology change was modeled by removing elements in contact with the virtual scalpel [23], as implemented in the SofaCarving plugin. Fixed boundary conditions were applied at the top and bottom surfaces, as well as near the patellar tendon, and lateral- and medial collateral ligaments. The carving distance and material stiffness were tuned by trial and error.

The characteristic force curve of the arthroscopic portal creation was modeled by performing linear interpolation in the measured force signal magnitude, as shown in Fig. 3. Because the material removal simulation using CFEM does not produce meaningful contact forces during cutting, the initial direction of the contact force (before material removal) was used for the haptic forces (see (21)). This collision detection was also used to trigger the computation of the haptic force. Because there was no meaningful simulation contact force, λ, signal fusion was omitted and the reference force scaled to fit the force capacity of the haptic device was used.

The resulting reference signal magnitude, haptic force signal magnitude, as well as force components of the haptic signal in the global simulation reference frame, are shown in Fig. 6. The simulation speed for the physics thread was about 25 Hz.

C. Application to Meniscus Examination

The characteristic force-displacement curve of meniscus examination was modelled as a nonlinear quadratic function following the characteristics shown in Fig. 5. The model takes in the magnitude of the linear-elastic force signal, $|\lambda|$, from the co-rotational finite element simulation, which uses the identified elastic modulus, E, as described in section IV-D2. We define our

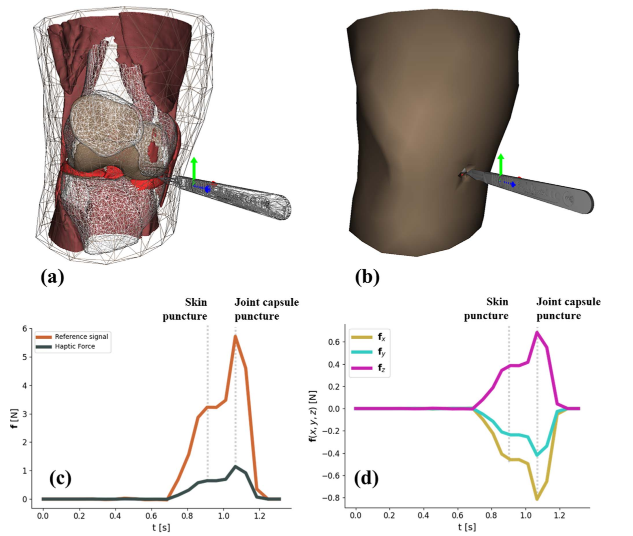

Fig. 6. (a) Wireframe view of the knee model used in the arthroscopic portal creation simulation. (b) Shaded view of the knee model. This is the user view. (c) Simulated reference signal magnitude and scaled haptic force signal magnitude. The skin puncture and joint capsule puncture locations are indicated. (d) The resulting force components of the haptic force signal in the global simulation reference frame.

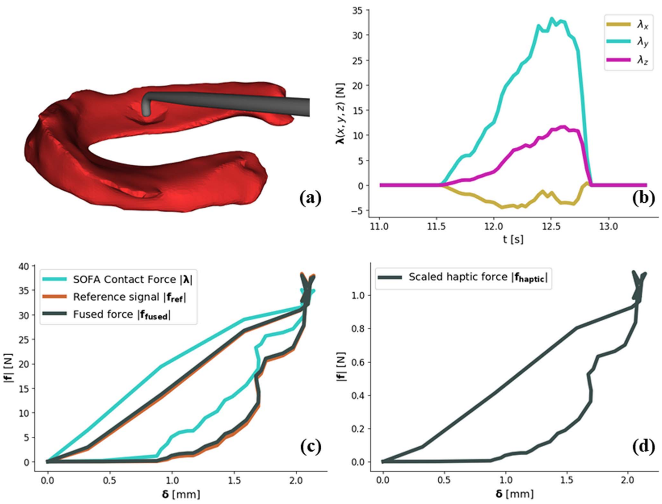

Fig. 7. (a) Visualization of the contact between probe and meniscus in SOFA. The elastic modulus of the meniscus was set to $E = 3.18$ MPa. (b) The resulting contact forces, $\lambda(x, y, z)$, from mechanical interaction between the probe and meniscus tissue. (c) The SOFA contact force magnitude, $|\lambda|$, compared to the nonlinear reference signal, $|\mathbf{f_{ref}}|$, and fused force signal, $|\mathbf{f_{haptic}}|$. (d) The haptic signal scaled to the force range of the haptic device.

nonlinear force signal to be:

$$|\mathbf{f_{ref}}| = \beta|\lambda|^2, \tag{23}$$

where β is defined as $\beta = \frac{1}{f_{\text{peak}}}$, and f_{peak} is the force where $|\mathbf{f_{ref}}| = |\lambda|$. Ideally, we let $f_{\text{peak}} = 10.77\,\text{N}$ as shown in Table II. However, for our demonstration, the haptic device is limited to a maximum force of 3 N. Therefore, the user will apply a larger deformation than expected because of the low force feedback, and the computed virtual contact force, λ, will become correspondingly higher. Thus the force where $|\mathbf{f_{ref}}| = |\lambda|$ needs to be set higher. Mapping the force range $(0\,\text{N}, 10.77\,\text{N}) \mapsto (0\,\text{N}, 3\,\text{N})$, we let $f_{\text{peak}} = 10.77\,\text{N} \cdot 3 = 32.31\,\text{N}$. The reference force signal, $|\mathbf{f_{ref}}|$, was then fused with λ using the Kalman filter, as described in Section III-D, with $R_\lambda = 0.001$ $R_{\mathbf{ref}} = 0.774$. Finally, haptic force signal magnitude, $|\mathbf{f_{haptic}}|$, is scaled to fit the force range of the haptic device to ensure stable haptic display. The achieved refresh rate for the physics thread was about 47 Hz during continuous contact. This effectively computes a nonlinear characteristic force signal based on a linear-elastic modulus. An example of resulting force signals for elastic modulus, $E = 3.18\,\text{MPa}$, is shown in Fig. 7.

VI. User Validation Experiments: Construct and Face Validity

A. Overview

This section describes user experiments that evaluate our reality-based method compared to linear-elasticity based kinesthetic feedback. We compare the performance of experts and novices in an arthroscopic portal creation and a meniscus examination task with respect to construct validity. Construct validity refers to discriminating between novice and expert performance, and assumes that if the expert perform better than the novice, the simulation reflects skills that resemble surgical skills [3]. Finally, some aspects on face validity, which is a subjective qualitative measure of how well the simulation represents actual surgery, is included.

B. Study Design

Handling of personal data according to GDPR was considered by the Norwegian Agency for Shared Services in Education and Research (Sikt) under reference 599785. All participant data was made anonymous. Informed consent was obtained from all participants.

24 participants were recruited to participate in the study, of which 12 (10 males, 2 females) were experienced orthopedic surgeons (> 1 year arthroscopy experience), and 12 (8 males, 4 females) had no prior surgical experience. Experts were recruited from two different hospitals. The mean age was 43.6 years in the expert group and 37.2 years in the novice group. For the experts, the mean experience with arthroscopic surgery was 11.8 years. The experiment procedure consisted of three main phases: familiarization, meniscus examination, and knee portal creation. The setup is shown in Fig. 8.

1) Familiarization: The purpose of the familiarization phase was to reduce familiarization bias. Participants were given instructions on how to operate the haptic device and interact with tissue in the virtual environment. They were then invited to explore a knee model consisting of rigid bones (femur, patella, tibia, fibula), deformable meniscus tissue (lateral and medial), deformable ligaments (ACL and PCL), and visual models (LCL, PCL, muscle). Participants were encouraged to feel the difference between rigid and deformable objects.

2) Meniscus Examination: The purpose of the meniscus examination phase was to distinguish between various levels of meniscus stiffness. This exercise is clinically relevant because

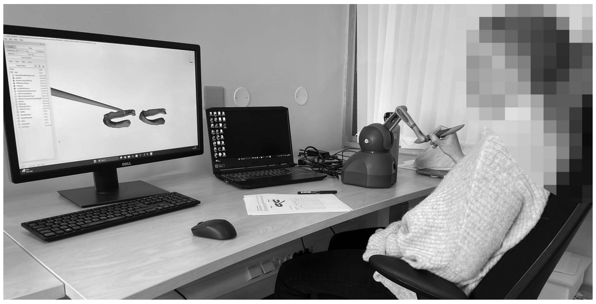

Fig. 8. User experiment setup with Geomagic Touch haptic device and simulation scenes developed in SOFA v.22.06. Participants filled out a paper survey during the experiment.

TABLE III
DURING PHASE 2), PARTICIPANTS WERE ASKED TO IDENTIFY WHETHER THEY PERCEIVED THE HIGHER OR LOWER MODULUS AS THE STIFFEST.

Difficulty	Lower Modulus	Higher Modulus	Difference
Level 1	1.41 MPa	5.05 MPa	3.64 MPa
Level 2	1.41 MPa	4.24 MPa	2.83 MPa
	2.67 MPa	4.24 MPa	2.38 MPa
Level 3	2.67 MPa	4.24 MPa	1.57 MPa
Level 4	4.24 MPa	5.05 MPa	0.81 MPa
	3.85 MPa	4.24 MPa	0.39 MPa
Level 5	3.85 MPa	3.85 MPa	0 MPa
	5.05 MPa	5.05 MPa	0 MPa

The presentation order was randomized. The five different stiffness levels were determined based on differences in modulus.

it represents distinguishing between healthy and degenerative meniscus tissue in diagnostic knee arthroscopy. The difference threshold (DT) was used as a metric, and established for both the linear-elastic and reality-based methods. The DT indicates a point at which participants can no longer reliably distinguish between stiffness levels, and was taken as the midpoint of a psychometric function spanning from chance to perfect performance, according to [38]. Following construct validity, we hypothesize that experts have finer sensitivity than novices for both methods. For this probing task, we emphasize that the goal is not to improve sensitivity but to improve realism. We therefore use an exercise requiring fine sensitivity to show that the reality-based method can achieve the same sensitivity as a linear-elastic method, which is considered the gold standard. Thus, we hypothesize no difference in sensitivity between the two methods.

Five different elastic moduli were included based on our ex vivo modulus measurements as well as reported decrease in modulus based on degenerative arthrosis development [6], [33]. The moduli were 5.05 MPa, 4.24 MPa, 3.85 MPa, 2.67 MPa and 1.41 MPa. The moduli were combined to form five different difficulty levels based on differences in elastic moduli. A higher difficulty was associated with a smaller difference in modulus. An overview is shown in Table III.

Participants were shown two visually identical lateral menisci in the simulation environment and asked to report in a paper survey which meniscus they perceived as the stiffest. The test design was a three-option forced choice, with options being meniscus A (left), meniscus B (right), or equally stiff. The test was repeated ten times. In each participant group (experts and novices), half of the participants experienced reality-based kinesthetic feedback (see section V-C), and the other half was from linear-elastic based kinesthetic feedback. User force, displacement, and decision time were automatically recorded in the software. After the ten tests, experts were invited to fill out a paper survey regarding how similar the simulation was to the actual procedure. The survey included Likert-scale questions rating haptic feedback and visual representation on a scale from one to five, as well as free-text answers.

TABLE IV
STATISTICAL COMPARISON OF DT USING MANN-WHITNEY U TEST.

	DL	Novices			Experts		
		μ	U	p	μ	U	p
Linear-elastic haptics	2	91.67%	21.0	0.69	91.67%	21.0	0.69
	3	33.33%	30.0	0.06	100%	18.0	1.0
	4	33.33%	36.0	<0.05	41.67%	33.0	<0.05
	5	59.33%	27.0	0.18	59.33%	33.0	<0.05
Reality-based haptics	2	100%	18.0	1.0	91.67%	18.0	1.0
	3	83.3%	21.0	0.69	83.3%	18.5	1.0
	4	33.3%	36.0	<0.05	33.3%	31.5	<0.05
	5	58.3%	33.0	<0.05	66.7%	24.5	0.39

All difficulty levels (DL) are compared to level 1. The mean score, μ, is reported as a descriptive statistic. However, we emphasize that the median is used in the statistical comparison. Results show that none of the groups could detect 0.81 mpa difference in stiffness.

3) Knee Portal Creation: The purpose of the third task was to practice knee portal creation using a scalpel. For this procedure, kinesthetic feedback is missing in existing commercial arthroscopy simulators. Following construct validity, we hypothesize that experts insert the scalpel a shorter distance than novices after puncture occurs.

Participants were asked to create one medial and one lateral arthroscopic portal in a knee model using a virtual scalpel (see section V-B). The instructor explained that the task was to insert the scalpel as a stab incision and that the users would feel a 'pop' when the joint capsule was punctured. Further, it was explained that the participants would need to apply enough force to puncture the joint capsule, but take care not to damage internal structures. The procedure was repeated three times, but only the last two experiments were included to reduce familiarity bias. Scalpel insertion length was automatically measured using the software. Experts were invited to fill out a paper survey with Likert-scale questions and free-text answers regarding the realism of the simulation after completion.

C. Results

1) Meniscus Examination Difference Threshold: Statistical analysis of the survey data was performed in Python (v3.11.2) using the Scipy statistics module. The data failed to meet the normality assumption upon visual histogram inspection. The Mann-Whitney U test was therefore chosen to compare data. The mean identification score for all four groups was 95.8% for level 1 (3.64 MPa difference in stiffness). The groups were linear-elastic novice (LN), reality-based novice (RN), linear-elastic expert (LE), and reality-based expert (RE). As such, level 1 was used as a reference of comparison with the other difficulty levels.

The results are shown in Table IV. All groups could identify level 2 with more than 90% accuracy. For level 3, experts performed better than novices in the linear group, but the difference

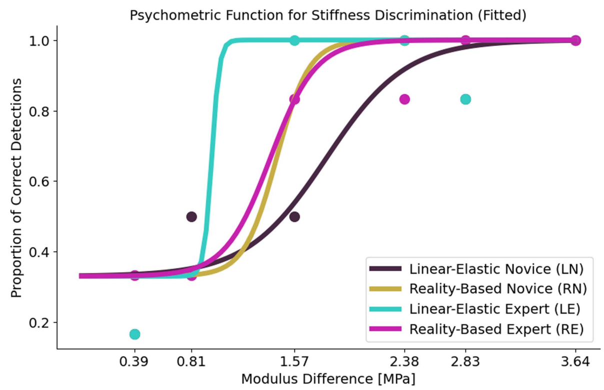

Fig. 9. Psychometric functions fitted to survey data. The LE group had the lowest DT (highest sensitivity), and the LN group had the highest DT (lowest sensitivity).

TABLE V
DESCRIPTIVE STATISTICS FOR DECISION TIME, MEAN VIRTUAL CONTACT FORCE AND MAXIMUM DISPLACEMENT FOR THE GROUPS EXPERT LINEAR-ELASTIC (LE), EXPERT REALITY-BASED (RE), NOVICE LINEAR-ELASTIC (LN) AND NOVICE REALITY-BASED (RN)

	Decision Time	**Mean Force**	**Max Displacement**
	μ	μ	μ
LE	63.9 s	17.4 N	3.2 mm
RE	52.5 s	21.1 N	3.3 mm
LN	91.6 s	42.1 N	3.2 mm
RN	70.8 s	34.8 N	2.2 mm

Experts had shorter decision times and used less force than novices.

for novices was not significant ($p = 0.06, U = 30.0$). There was a significant difference between level 4 and level 1 for all groups ($p < 0.05, U = 36.0; 36.0; 33.0; 31.5$). This indicates that no groups could distinguish between differences in modulus less than 0.81 MPa. The difference threshold was therefore higher than 0.81 MPa for all groups. For no difference in stiffness, the mean score was higher than level 4 for all groups. This indicates that it was easier to identify equal stiffness than small differences.

A psychometric Sigmoid function for each group was fitted based on the data according to [38]. The curves are shown in Fig. 9. The lower asymptote represents the probability of a correct answer by chance from the three-option forced choice survey. The difference threshold (DT) was found to be 1.80 MPa for LN, 1.47 MPa for RN, 0.99 MPa for LE, and 1.39 MPa for RE. A lower DT (finer sensitivity) for experts in both the linear-elastic and reality-based groups indicates construct validity.

2) Meniscus Examination - Decision Time, Force Use and Deformation: Decision times, mean force, and maximum probe displacements were automatically measured during the experiments. The decision time was defined as the time from first contact until the end. Statistical analysis was performed to compare groups LN, RN, LE, and RE. The data did not exhibit deviance from normality by visual inspection. A two-way ANOVA test in Python was therefore used.

Descriptive results are shown in Table V. Results show that experts had significantly shorter decision times than novices ($F_{(1,196)} = 12.34, p < 0.05, \eta^2_{\text{partial}} = 0.06$), and that experts used significantly less force ($F_{(1,159748)} = 141.88, p < 0.05, \eta^2_{\text{partial}} < 0.01$). This supports construct validity.

Further, the decision times were significantly shorter for reality-based haptics compared to linear-elastic haptics ($F_{(1,196)} = 6.12, p < 0.05, \eta^2_{\text{partial}} = 0.03$). For max displacements, reality-based groups (combined) had significantly smaller probe displacements than the linear-elastic group ($F_{(1,31276)} = 298.99, p < 0.05, \eta^2_{\text{partial}} = 0.01$). However, this was primarily due to the novice group, and this trend was not apparent in the expert group.

3) Meniscus Examination - Face Validity: A qualitative assessment of the face validity was performed from Likert survey results and free text answers in the expert group. From the five-point Likert survey, the linear-elastic based haptic feedback obtained a mean score of 4.0, and the mean score for the reality-based haptic feedback was 3.4. For the visual representation, the scores were respectively 3.3 and 4.0 in the linear-elastic and reality-based groups. For the free text answers, four experts commented that missed collision detection with "pop-through" was disturbing (both LE and RB groups). Three commented that there was too much visual deformation (both groups). Two reported that visual depth perception was challenging, and several others also commented on this to the instructor. Five in the LE group and four in the RB group reported that the haptic feedback was very good. Two in each group reported that the haptic feedback was realistic.

4) Portal Creation - Scalpel Insertion Length: For knee portal creation, the scalpel insertion length was statistically compared for experts and novices. The data did not exhibit deviance from normality by visual inspection, and a paired T-test was therefore used for comparison. The mean insertion length was 19.91 ± 15.43 mm for the novice group, and 21.25 ± 15.03 mm for the expert group. There was no statistically significant difference between the groups ($p = 0.85$). The length of the virtual scalpel blade was 40 mm.

5) Portal Creation - Face Validity: A qualitative assessment of face validity for portal creation was performed from the five-point Likert survey and free text answers from experts. The haptic feedback obtained a mean score of 2.4, and the visual representation obtained 2.3. From the free text answers, eight out of twelve experts reported that they struggled with sense of depth. Three reported that visual cuts removed too much material. For the haptic feedback, three reported that the joint capsule puncture felt clear, but the skin puncture was less distinct. The general feedback was that excessive visual tissue removal dominated the overall impression, but haptic feedback functioned well on its own.

VII. DISCUSSION

Considering the meniscus examination task in the user validation study, which represents general soft tissue contact, the findings indicate that the difference threshold was lower for experts than novices in both the linear-elastic and reality-based groups. This indicates a finer sensitivity for experts. Also, experts used less force and had shorter decision times than novices. This supports construct validity for both methods.

When comparing haptic methods for meniscus examination, experts had slightly finer sensitivity using the linear-elastic method (0.99 MPa) than the reality-based method (1.39 MPa). In contrast, novices had slightly finer sensitivity with the reality-based method (1.47 MPa) than the linear-elastic method (1.80 MPa). From this one might hypothesize that the proposed reality-based method confers advantages to novices compared to experts. However, qualitative feedback did not provide evidence to support this. In addition, the Mann-Whitney test did not reveal statistical differences between the methods. The aim was to show that the reality-based method provides stable nonlinear kinesthetic feedback matching the sensitivity of the linear-elastic method, not to enhance stiffness discrimination sensitivity. A limitation of the user study design was that it focused on experts and novices and did not consider intermediate levels. Integrating participants with intermediate skill levels, such as junior resident physicians, could potentially reveal differences in sensitivity between methods.

The arthroscopic portal creation task demonstrated continuous kinesthetic feedback during soft tissue removal. Here, the difference between experts and novices was not statistically significant ($p = 0.85$). This could be attributed to a coarse finite element mesh which led to minor inconsistencies between visual and haptic display during tissue removal. Notably, visual perception has been shown to dominate haptic perception, and inconsistencies should be carefully avoided [39]. One limitation of this method is that visual material removal depends on how far the instrument penetrates the tissue, and tissue stiffness and boundary conditions also affect instrument travel. While it is possible to adjust this factor to align visual and haptic feedback, a more effective approach might be to trigger visual material removal directly based on haptic force limits. Alternatively, a finer mesh would likely yield better results. As such, user validation was inconclusive in demonstrating construct validity for the entire simulation setup (visual and haptic). However, haptic feedback functioned well on its own, with kinesthetic force curves consistent with experimental measurements, and without reported user issues.

The simulation setup was limited by the force capacity of the haptic device (3 N) and the refresh rate of the physics thread. This was apparent from qualitative feedback from the expert group. One participant (male, 42 years old) explained there was more visual deformation in meniscus examination than expected, which was a direct consequence of the force capacity and stability of the haptic device. Similar comments were received from experts in both the linear-elasticity and reality-based groups, albeit the deformation was significantly smaller ($p < 0.05$) for the reality-based method. The mean maximum displacement was 3.3 mm in the linear-elasticity groups (combined), and 2.7 mm for the reality-based groups (combined). Using the reality-based method, visual deformation can be further limited by setting f_{peak} to a smaller value. However, this requires higher refresh rates to avoid stability issues [10]. It is likely that a haptic device with a higher force capacity would improve the face validity of the simulation. With a mean peak force of 10.77 N in meniscus probing experiments, only three commercial kinesthetic devices meet this force requirement according to Haptipedia [40]. These are the Force Dimension Omega, Phantom Premium 1.5 high force, and Haption Virtuose 6D high force. However, considering functional task alignment of arthroscopic tasks, such as learning tactile sensations and dexterous motor skills of specialized surgical equipment, surgery-like devices, such as Follou Haptics could be equally important.

Considering the computational cost of the proposed reality-based method, there was only a minor reduction in speed. For the meniscus probing task, the speed was about 80 Hz when the probe was moving in free space. During continuous contact using only the linear-elastic force feedback, the speed was reduced to about 52 Hz. Further, by introducing contact force extraction in Python (see (7)), the speed was reduced to about 47 Hz. Finally, by introducing nonlinear force computation (23), Kalman filter sensor fusion (Algorithm 1) and transmitting forces to haptic device (20), the speed remained at about 47 Hz. It shall be noted that contact force extraction, which represented the majority of the speed reduction, was implemented in Python, and that C++ implementation could minimize this speed reduction. This clearly demonstrates that our method did not introduce a substantial loss of computation speed.

A limitation of this study is that it did not consider multimodal haptics, for example by including vibrotactile display of contact and slip vibrations. Other studies have demonstrated an improvement in perceived realism by displaying transient vibrations in combination with kinesthetic feedback during object contact, for example during tapping [41], sliding and puncture [42], and bone drilling [28]. These vibrations can be conveyed via the motors of the kinesthetic device and can be seamlessly incorporated into the proposed method. One challenge of the data collection setup in this paper is maintaining an adequate sampling rate for such vibrations.

A challenge in adopting the proposed reality-based method could be system integration. To mitigate this, we have made a Python implementation openly available on GitHub [43]. Simulation scenes used in the user validation study are also available.

For future work, the method should be evaluated for characteristic force signals from other surgical maneuvers such as shaving or suturing the meniscus, and from other methods, such as neural network models. Also, exploration of the method within the position-based dynamics framework is interesting because higher refresh rates can be combined with reality-based force signals, thereby simultaneously addressing limitations in computation speed and characteristics of the haptic force.

VIII. Conclusion

This paper presented a reality-based method for the enrichment of kinesthetic haptic rendering in surgical simulations. The method was compared to linear-elastic kinesthetic feedback in arthroscopic meniscus examination through a user experiment with experts and novices. The difference thresholds were found to be 1.80 MPa (novice, linear-elastic), 1.47 MPa (novice, reality-based), 0.99 MPa (expert, linear-elastic), and 1.39 MPa (expert, reality-based). Experts used significantly less force and had shorter decision times ($p < 0.05$), supporting construct validity for both haptic feedback approaches. We remark that a well-tuned linear elastic simulation could be sufficient for meniscus stiffness discrimination, but is limited to a linear force response. The reality-based method also demonstrated continuous kinesthetic feedback free of instabilities during arthroscopic portal creation, but user validation was inconclusive due to minor inconsistencies with visual and haptic integration. The benefit of the presented method is that it simplifies procedure-specific haptic rendering in surgical simulation development.

Acknowledgment

The authors would like to thank Hugo Talbot, Paul Baksic, and the SOFA consortium for support with simulator development. The authors would also like to thank Alf Inge Hellevik and Eivind Inderhaug for help with organizing user experiments, and the volunteers who participated. Carl Petter Kulseng and Kjell-Inge Gjesdal at NordicCAD are recognized for providing the anatomical models used in this project.

References

[1] S. Chae, S.-W. Jung, and H.-S. Park, "In vivo biomechanical measurement and haptic simulation of portal placement procedure in shoulder arthroscopic surgery," *PLoS One*, vol. 13, no. 3, Mar. 2018, Art. no. e0193736.

[2] Y. Hui, O. Safir, A. Dubrowski, and H. Carnahan, "What skills should simulation training in arthroscopy teach residents? A focus on resident input," *Int. J. Comput. Assist. Radiol. Surg.*, vol. 8, no. 6, pp. 945–953, Nov. 2013.

[3] R. M. Frank, "Editorial commentary: Arthroscopic simulators—Are we there yet?," *Arthroscopy: J. Arthroscopic Related Surg.*, vol. 35, no. 8, pp. 2391–2393, Aug. 2019.

[4] E. M. Overtoom, T. Horeman, F.-W. Jansen, J. Dankelman, and H. W. R. Schreuder, "Haptic feedback, force feedback, and force-sensing in simulation training for laparoscopy: A systematic overview," *J. Surg. Educ.*, vol. 76, no. 1, pp. 242–261, Jan. 2019.

[5] C. Pacchierotti and D. Prattichizzo, "Cutaneous/Tactile haptic feedback in robotic teleoperation: Motivation, survey, and perspectives," *IEEE Trans. Robot.*, vol. 40, pp. 978–998, 2024.

[6] K. M. Fischenich, J. Lewis, K. A. Kindsfater, T. S. Bailey, and T. L. Haut Donahue, "Effects of degeneration on the compressive and tensile properties of human meniscus," *J. Biomech.*, vol. 48, no. 8, pp. 1407–1411, Jun. 2015.

[7] U. Erden, M. A. Gromski, S. De, and D. Demirel, "Preliminary validation of the virtual bariatric endoscopic simulator," *iGIE*, vol. 3, no. 4, pp. 453–462, Dec. 2024.

[8] J. Bender, M. Müller, and M. Macklin, "Position-based simulation methods in computer graphics," in *EUROGRAPHICS Tutorials*, Zürich, Switzerland: The Eurographics Association, pp. 1–32, 2015.

[9] M. Freutel, H. Schmidt, L. Dürselen, A. Ignatius, and F. Galbusera, "Finite element modeling of soft tissues: Material models, tissue interaction and challenges," *Clin. Biomech.*, vol. 29, no. 4, pp. 363–372, 2014.

[10] J. Colgate, M. Stanley, and J. Brown, "Issues in the haptic display of tool use," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. Hum. Robot Interact. Cooperative Robots*, vol. 3, Aug. 1995, pp. 140–145.

[11] B. Hannaford and A. M. Okamura, "Haptics," in *Springer Handbook of Robotics*, B. Siciliano and O. Khatib, Eds. Berlin, Germany: Springer, 2008, pp. 719–739.

[12] C. Zilles and J. Salisbury, "A constraint-based God-object method for haptic display," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. Hum. Robot Interact. Cooperative Robots*, vol. 3, Aug. 1995, pp. 146–151.

[13] C. Duriez, F. Dubois, A. Kheddar, and C. Andriot, "Realistic haptic rendering of interacting deformable objects in virtual environments," *IEEE Trans. Vis. Comput. Graph.*, vol. 12, no. 1, pp. 36–47, Jan./Feb. 2006.

[14] I. Peterlik, M. Nouicer, C. Duriez, S. Cotin, and A. Kheddar, "Constraint-based haptic rendering of multirate compliant mechanisms," *IEEE Trans. Haptics*, vol. 4, no. 3, pp. 175–187, Jul./Sep. 2011.

[15] F. Faure et al., "SOFA: A multi-model framework for interactive physical simulation," *Soft Tissue Biomech. Model. Comput. Assist. Surg.*, 2012, vol. 11, pp. 283–321.

[16] O. Bjelland et al., "Toward a digital twin for arthroscopic knee surgery: A systematic review," *IEEE Access*, vol. 10, pp. 45029–45052, 2022.

[17] A. C. Abraham, C. R. Edwards, G. M. Odegard, and T. L. H. Donahue, "Regional and fiber orientation dependent shear properties and anisotropy of bovine meniscus," *J. Mech. Behav. Biomed. Mater.*, vol. 4, no. 8, pp. 2024–2030, Nov. 2011.

[18] P. Shull, K. Bark, and M. Cutkosky, "Skin nonlinearities and their effect on user perception for rotational skin stretch," in *Proc. IEEE Haptics Symp.*, Mar. 2010, pp. 77–82.

[19] B. Smith, F. D. Goes, and T. Kim, "Stable Neo-Hookean flesh simulation," *ACM Trans. Graph.*, vol. 37, no. 2, pp. 12:1–12:15, Mar. 2018.

[20] N. Schulmann, S. Cotin, and I. Peterlik, "The effect of discretization on parameter identification. Application to patient-specific simulations," in *Proc. Comput. Methods, Imag. Visual. Biomech. Biomed. Eng.*, 2020, pp. 237–247.

[21] M. Kaluschke, R. Weller, N. Hammer, L. Pelliccia, M. Lorenz, and G. Zachmann, "Realistic haptic feedback for material removal in medical simulations," in *Proc. IEEE Haptics Symp.*, Mar. 2020, pp. 920–926.

[22] H. Courtecuisse, H. Jung, J. Allard, C. Duriez, D. Y. Lee, and S. Cotin, "GPU-based real-time soft tissue deformation with cutting and haptic feedback," *Prog. Biophys. Mol. Biol.*, vol. 103, no. 2, pp. 159–168, Dec. 2010.

[23] M. Bro-Nielsen, "Finite element modeling in surgery simulation," in *Proc. IEEE*, vol. 86, no. 3, pp. 490–503, Mar. 1998.

[24] E. Burman, S. Claus, P. Hansbo, M. G. Larson, and A. Massing, "CutFEM: Discretizing geometry and partial differential equations," *Int. J. Numer. Methods Eng.*, vol. 104, no. 7, pp. 472–501, 2015.

[25] H. P. Bui, S. Tomar, and S. P. A. Bordas, "Corotational cut finite element method for real-time surgical simulation: Application to needle insertion simulation," *Comput. Methods Appl. Mechan. Eng.*, vol. 345, pp. 183–211, Mar. 2019.

[26] A. Okamura, M. Cutkosky, and J. Dennerlein, "Reality-based models for vibration feedback in virtual environments," *IEEE/ASME Trans. Mechatron.*, vol. 6, no. 3, pp. 245–252, Sep. 2001.

[27] A. Okamura, C. Simone, and M. O'Leary, "Force modeling for needle insertion into soft tissue," *IEEE Trans. Biomed. Eng.*, vol. 51, no. 10, pp. 1707–1716, Oct. 2004.

[28] Q. Ha-Van, H. Schwendinger, Y. Kim, and M. Harders, "Design and characterization of an actuated drill mockup for orthopedic surgical training," *IEEE Trans. Haptics*, vol. 13, no. 4, pp. 655–667, Oct. 2020.

[29] A. Galvan, A. K. Da Costa, J. Shields, K. Kho, and A. M. Fey, "Haptic simulator for trocar insertion training," in *Proc. IEEE World Haptics Conf.*, Jul. 2021, pp. 397–402.

[30] O. Bjelland, M. D. Pedersen, M. Steinert, and R. T. Bye, "Intraoperative data-based haptic feedback for arthroscopic partial meniscectomy punch simulation," *IEEE Access*, vol. 10, pp. 107269–107282, 2022.

[31] F. Picard, A. H. Deakin, P. E. Riches, K. Deep, and J. Baines, "Computer assisted orthopaedic surgery: Past, present and future," *Med. Eng. Phys.*, vol. 72, pp. 55–65, Oct. 2019.

[32] J. B. Joolee and S. Jeon, "Data-driven haptic texture modeling and rendering based on deep spatio-temporal networks," *IEEE Trans. Haptics*, vol. 15, no. 1, pp. 62–67, Jan. 2022.

[33] B. Rasheed et al., "Intraoperative identification of patient-specific elastic modulus of the meniscus during arthroscopy," *Comput. Methods Programs Biomed.*, vol. 254, Sep. 2024, Art. no. 108269.

[34] O. Bjelland, L. I. Hatledal, R. T. Bye, and M. Steinert, "Implementation and evaluation of an arthroscopic tracker system for intraoperative motion tracking and force registration," in *Proc. 37th ECMS Int. Conf. Model. Simul.*, Jun. 2023, pp. 459–465.

[35] G. Welch and G. Bishop, "An introduction to the Kalman filter," University of North Carolina at Chapel Hill, White Paper, 1997. [Online]. Available: https://perso.crans.org/club-krobot/doc/kalman.pdf

[36] C. P. S. Kulseng, V. Nainamalai, E. Grøvik, J.-T. Geitung, A. Årøen, and K.-I. Gjesdal, "Automatic segmentation of human knee anatomy by a convolutional neural network applying a 3D MRI protocol," *BMC Musculoskelet. Disord.*, vol. 24, no. 1, Jan. 2023, Art. no. 41.

[37] C. Geuzaine and J.-F. Remacle, "GMSH: A three-dimensional finite element mesh generator with built-in pre- and post-processing facilities," *Int. J. Numer. Methods Eng.* 79, no. 11, pp. 1309–1331, 2009.

[38] L. A. Jones and H. Z. Tan, "Application of psychophysical techniques to haptic research," *IEEE Trans. Haptics*, vol. 6, no. 3, pp. 268–284, Jul.–Sep. 2013.

[39] Y. Xu, S. O'Keefe, S. Suzuki, and S. L. Franconeri, "Visual influence on haptic torque perception," *Perception*, vol. 41, no. 7, pp. 862–870, Jul. 2012.

[40] H. Seifi et al., "Haptipedia: Accelerating haptic device discovery to support interaction & engineering design," in *Proc. CHI Conf. Hum. Factors Comput. Syst.*, May 2019, pp. 1–12.
[41] K. J. Kuchenbecker, J. Fiene, and G. Niemeyer, "Improving contact realism through event-based haptic feedback," *IEEE Trans. Vis. Comput. Graph.*, vol. 12, no. 2, pp. 219–230, Mar./Apr. 2006.
[42] A. Okamura, J. Dennerlein, and R. Howe, "Vibration feedback models for virtual environments," in *Proc. IEEE Int. Conf. Robot. Automat.*, vol. 1, May 1998, pp. 674–679.
[43] B. Øystein, "Reality-based haptics for SOFA," Apr. 2025. [Online]. Available: https://github.com/CHAOS-AES/Reality-Based-Haptics-For-SOFA

Øystein Bjelland received the M.Sc. degree in mechanical engineering from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 2016, and the Ph.D. degree in engineering cybernetics from NTNU, Ålesund, Norway, in 2025. From 2016 to 2020, he was a Mechanical Design Engineer of applied research and development with Prototech AS, working with Subsea Robotics, Flow Measurement Technology, Fuel Cell Systems for Space Applications, and Metal Additive Manufacturing. Since 2020, he has been a Researcher with the Cyber-Physical Systems Lab, NTNU, and is also with Aalesund Biomechanics Laboratory and TrollLabs. His research interests include digital twins, mechatronic systems, and biomechanics, and focuses on advancing haptic technology for interactive surgical training simulations in orthopedic applications.

Bismi Rasheed graduated in civil engineering from the University of Calicut, Malappuram, Kerala, India, in 2014, the M.Tech degree in computer aided structural analysis and design from the Cochin University of Science and Technology, Kochi, Kerala, India, in 2017. She is currently working toward the Ph.D. degree with the Department of ICT and Natural Sciences at NTNU, Ålesund, Norway. From 2018 to 2020, she was a Research Scholar with the Department of Applied Mechanics, Indian Institute of Technology Madras, Chennai, India. She is also with the Cyber-Physical Systems Laboratory, NTNU Ålesund and Ålesund Biomechanics Laboratory. Her research interests include computational modeling, solid mechanics, and biomechanics. Current research interests focus on biomechanical characterization, soft tissue deformation modeling, and surgical simulation.

Intissar Cherif received the Bachelor of Science and master's degrees in computer science: Images and artificial life from the University of Mohamed Khider, Biskra, Algeria. She is currently working toward the Ph.D. degree in computer science with the University of Paris Saclay, IBISC Lab, France, in cotutelle with the University of Mohamed Khider, LESIA Lab, Algeria. Her research focuses on the Since 2024, she has been a Teaching and Research Assistant with the University of Évry - Paris Saclay, where she Teaches computer science and advanced programming while also supervising technology projects. Her research interests include enhancing training effectiveness in technical fields through immersive simulation technologies and fidelity of haptic feedback in virtual reality simulators for learning technical gestures. She was the recipient of the Eiffel Excellence Scholarship, awarded by the French Government to fund her ongoing Ph.D. degree.

Andreas Fagerhaug Dalen received the M.D. degree from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 2013. He is currently working toward the Ph.D. degree in orthopedics. He is also an Orthopedic Surgeon with Ålesund General Hospital. Since 2018, he has been the Head of Ålesund Biomechanics Laboratory (ÅBL), where his research is focused on ankle fractures and biomechanics. His research interests include biomechanics and treatment of injuries in shoulders, knees, and ankles.

Amine Chellali received the M.Sc. degree in robotics from the Ecole Centrale de Nantes, Nantes, France, in 2005, and the Ph.D. degree in HCI from the University of Nantes, Nantes, in 2009. From 2009 to 2011, he was a Research Assistant with the Research Institute on Communication and Cybernetics of Nantes (IRCCyN). From 2011 to 2012, he moved to the U.S. to work as a Postdoctoral Associate in ergonomics with Remote Environments Laboratory (EREL), Tufts University, Medford, MA, USA and then as a Research Associate with the Interactive Surgical Systems Laboratory (ISSyL), Harvard Medical School. He is currently an Associate Professor with the Department of Computer Engineering, University of Evry Paris-Saclay (France) and also a Co-Leader of the Interaction, Mixed Reality, and Assistive Robotics (IRA2) with IBISC Laboratory. His research interests include simulator fidelity, haptic interactions, collaborative virtual environments, and the transfer and evaluation of human skills, and main applications are in the field of surgical training. He was the recipient of the several grants for his research on VR simulation from the French National Research Agency.

Martin Steinert was born in Dresden, Germany. He received the B.A., M.A., and Ph.D. (Dr.rer.pol) degrees in technology management from the University of Fribourg, Fribourg, Switzerland. He has been an Assistant Professor with the University of Fribourg, Switzerland, Visiting Scholar with MIT, Cambridge, MA, USA, and Stanford University, Stanford, CA, USA, before changing full time to Stanford University as the Deputy Director with the Center for Design Research (CDR), and an Assistant Professor (Acting) of mechanical engineering with Stanford University. Since 2013, he has been a Full Professor of engineering design with the Department of Mechanical and Industrial Engineering (MTP), Norwegian University of Science and Technology (NTNU), Trondheim, Norway. His research interests include the fuzzy front end of new product development and design: optimizing the intersection of engineering design thinking and new product development, mechatronics/sensors, and computer sciences (especially machine learning). A special focus is on conceptual development and alpha prototype generation of high-performance requirements and on experimental tools and setups. He has more than 200 publications registered in Google Scholar, as of August 2021. Dr. Steinert was the recipient of the several prizes in both teaching and research. He has been a member of the Norwegian Academy of Technological Sciences (NTVA) since 2015.

Robin T. Bye (Senior Member, IEEE) was born in Ålesund, Norway, in 1979. He received the B.Eng. (with Hons.), M.Eng.Sc., and Ph.D. degrees in electrical engineering from the University of New South Wales, Sydney, Australia, in 2004, 2005, and 2009, respectively. Since 2008, he has been with the Department of ICT and Natural Sciences (IIR), Norwegian University of Science and Technology (NTNU), Ålesund, Norway, and became a Full Professor in 2020. He also Heads the Undergraduate Automation Engineering Programme and the Cyber-Physical Systems Laboratory. His research interests include the broad areas of cybernetics, artificial intelligence, neuroengineering, and engineering didactics. Prof. Bye was the recipient of the Goodeve Medal by The Operational Research Society in 2019.